Cornell Engineering
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EasiStitch™

MedThreads

DESIGN HISTORY FILE

Prepared for: Dr. Antaki, Dr. Cira, and Dr. Werth
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GENERIC GRADING RUBRIC (May be/Will be customized depending on the scope of individual projects. For example, not every project involves user testing, some may emphasize Engineering Standards more than others, etc.)

Executive Summary

Efficient and reliable wound closure is essential in emergency or surgical care because it minimizes the risk of postoperative infection, reduces overall recovery time, and improves cosmetic outcomes for patients. Traumatic wounds are common in emergency rooms and often involve uneven edges, foreign debris, and urgent blood loss. With around 11 million cases each year in the US, fast and safe closure methods are critical for proper healing (Nicks et al., 2010).

Current methods consist of sutures, staples, and adhesives with each method having its own limitations. Even though sutures offer precision, suturing is a time-consuming process. Similarly, staples are fast and efficient but cause significant scarring, while adhesives are inadequate for deep wounds. Addressing the drawbacks of each method requires a solution that will balance speed, precision, and affordability while minimizing complications such as infection and wound dehiscence.

The EasiStitch project aims to revolutionize wound closure in surgical and emergency settings by providing a hybrid solution that combines the precision of traditional suturing with the speed of skin stapling. The device will address common challenges in current methods, ranging from scarring from staples to the time-intensive nature of sutures. By taking advantage of a curved needle and sewing mechanism, EasiStitch uses a gear train system connected to roller pins to guide the curved needle in a smooth, circular motion, automating the hand-suturing process. The device also features ergonomic grips and an easily actuated push-button, giving users better control and precision placing uniform sutures with minimal scarring.

Existing devices like Medtronic's Endostitch and Boston Scientific's Overstitch are effective for endoscopic use but are too complex and specialized for emergency wound closure. Other automated suturing tools improve speed but suffer from poor ergonomics, unreliable needle transfer, and user fatigue, and have consequently not been implemented broadly. Our design brings the precision of advanced suturing to emergency care with a simple intuitive device actuated with a single button. This offers fast and precise closure, reduced training needs, and greater ease of use not currently available in the wound closure market.

Our engineering analysis was performed separately on the three working prototypes of our device as the final product is still in development. Our electrical circuit was validated through measurements confirming safe and reliable operation. The motor drew 82mA under load (1.02W), and the transistor circuit ensured full saturation with minimal heat generation, eliminating the need for a heat sink even in extreme conditions (50˚C). For mechanical proof of principle, we built a 5X scaled "Works Like" model with full mechanical functionality using a hand crank. It demonstrated consistent suture uniformity (length), reliable puncture depth, and accurate needle entry angle. Users also found the ergonomic prototype, consisting of the non-functional outer housing and internal components, as intuitive than a standard skin stapler. All of these metrics support the technical feasibility of our solution.

Easistitch will pursue a 510(k) regulatory pathway as a novel Class II medical device. Next steps include final R&D to refine the needle actuation, validating the scaled-down model, and pursuing regulatory approvals. Actions further down the development cycle include developing partnerships with medical distributors that will sell the product at the price of $75 to hospitals and surgical centers. By targeting wound closure for emergency and surgical settings, Easistitch has a potential to reach approximately 300,000 units in the first year, Easistitch hopes to capture 10-15% of the market in 5 years due to limited direct competition and strong demand in low-resource environments.

Problem Statement (Motivation)

Wound closure is a critical part of trauma care, especially for lacerations to the extremities, one of the most frequent reasons for ER visits. According to HCUP data and published literature, traumatic lacerations account for over 11 million emergency department visits annually in the U.S., representing more than 5% of all ER cases (Barnes et al., 2021; Nicks et al., 2010). These wounds are often irregular, bleeding, and at high risk of infection, requiring immediate closure to stabilize the patient and support long-term healing. However, clinicians are often forced to choose between manual suturing, which is time-consuming but cosmetically superior, and stapling, which is faster but associated with more scarring and higher complication rates (Shantz et al., 2012). In overcrowded or under-resourced emergency settings, this trade-off can negatively impact outcomes—especially for pediatric or rural patients.

Our project, EasiStitch, builds upon a previous prototype to develop a single-use, ergonomic suturing device that combines the speed of stapling with the precision of suturing. Existing solutions such as staplers and adhesives often fall short in trauma cases involving deep, contaminated, or jagged lacerations (Johnson et al., 1981; Kathare & Shinde, 2019). Similarly, current semi-automated suturing tools on the market tend to be bulky, expensive, or not designed for emergency use (Ge et al., 2021; Yu et al., 2020). EasiStitch uses a curved needle guided by an internal roller and gear mechanism, enabling consistent suture placement even by less experienced users. By improving mechanical design and focusing on clinical usability, our device addresses both clinical performance and bioethical principles like beneficence, justice, and non-maleficence (Stanford University, 2019), aiming to reduce the burden on emergency staff and improve patient outcomes

Final Design

Design Configuration

"Works-Like" Prototype

For this functional prototype, EasiStitch is scaled up by a factor of 5 to show the device's mechanism of the needle puncturing the skin. The power source for this mechanism is input by the user through a hand crank. Upon rotation of the hand crank, a gear within a gear trio begins to rotate as it is connected to the same shaft as the hand crank. It is important to note that the gear trio is necessary to ensure proper directionality (the two outer gears in the trio must rotate in the same direction). This step is crucial as it is the largest driver for the directionality of the rollers which directly interface with the needle movement. This gear trio is connected to shafts which are held in place through acrylic frames. The two outer gears in the trio are connected to an actuating gear that is downstream. Both actuating gears then rotate a pair of rollers through a complimentary gear mechanism. Through these rollers, the needle is able to move 360° along the needle track. Once the needle is ejected from one pair of the rollers, it is recaptured within the other pair of rollers. The needle track sits within a PLA base that also houses the shafts in place. To ensure that the needle has space to rotate and puncture the skin, there is an acrylic frame that sits at the base of the device.

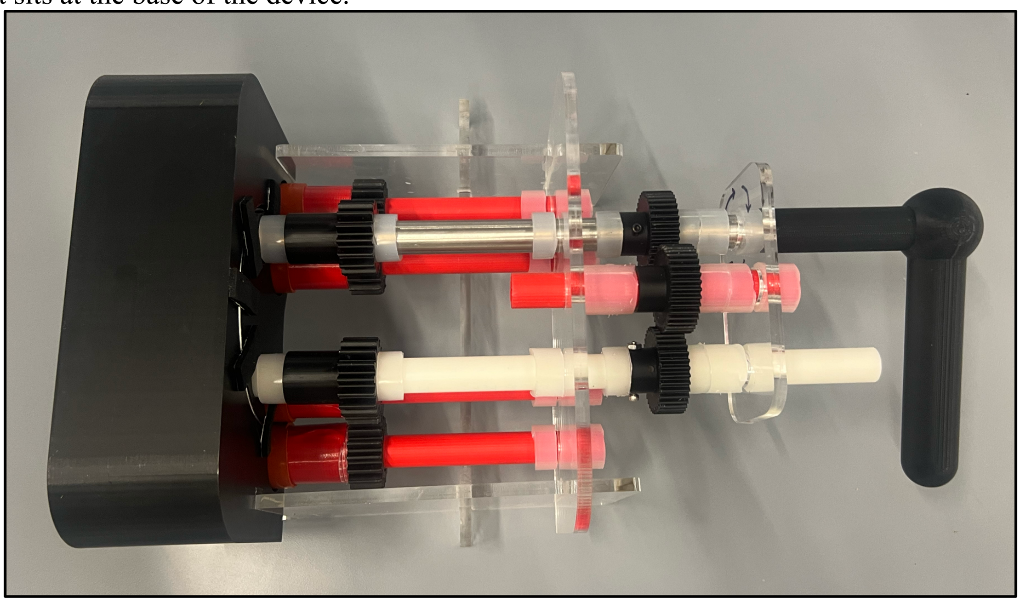

Figure 1. EasiStitch Works-Like Prototype

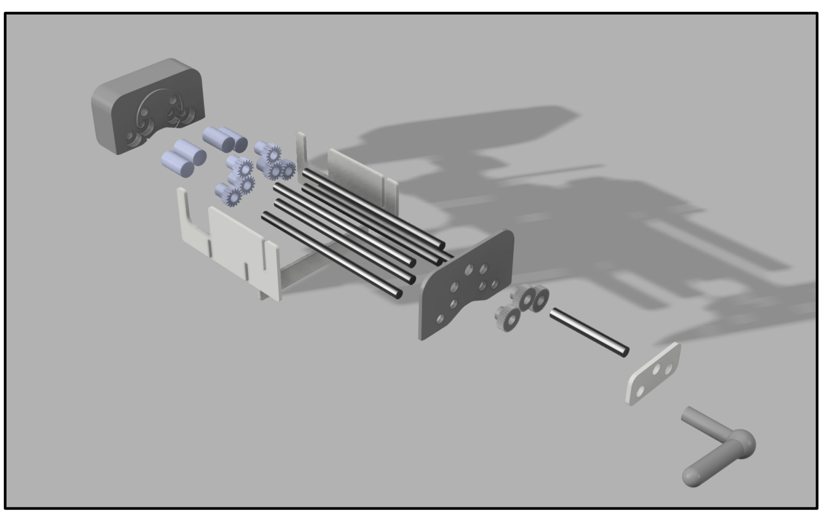

Figure 2. Exploded Rendering of Works-Like Model

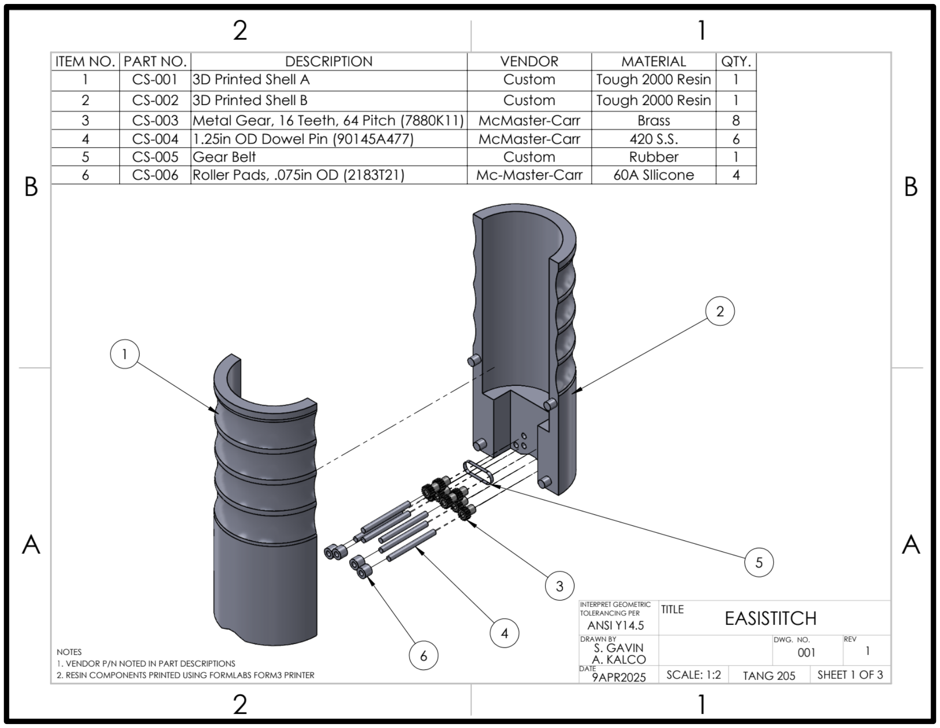

Figure 3. Ergonomic Model Assembly Drawing

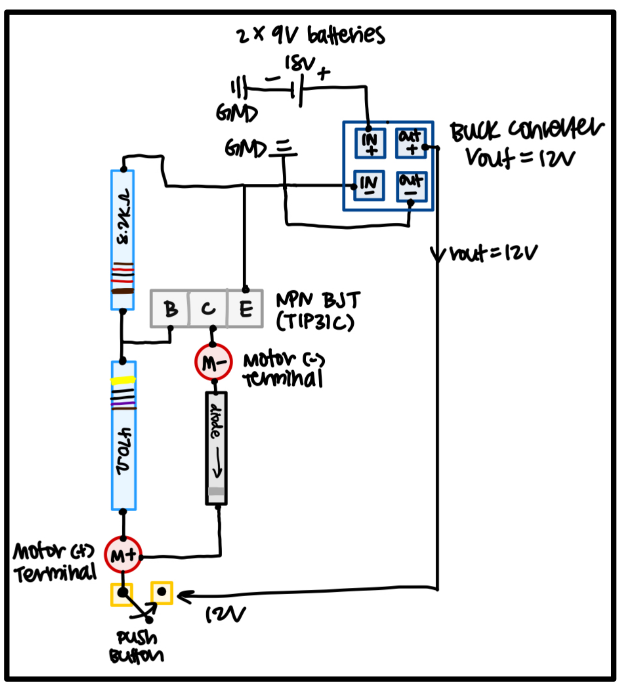

Figure 5. Internal Electrical Components

The motor is powered by two 9V batteries in series (18V) which are scaled down with a LM2596 module buck converter to 12V. The batteries and buck converter reside in a custom battery pack with a manual switch that powers on and off the device. The motor's positive terminal is connected to a pushbutton which supplies 12V to the motor when pressed. The negative terminal of the motor connects to the collector of the transistor. The emitter is connected to ground, completing the path for current flow. The base of the transistor is driven through a 470 ohm resistor from the motors positive terminal, and is also connected to ground through an 8.2 kohm pull-down resistor. When the pushbutton is pressed, the positive motor terminal node is at 12V, turning the transistor on by providing a base-emitter voltage of 12V, which is over the threshold (.7V) to turn the transistor on and allow current to flow between the collector and emitter such that current in the circuit flows from the 12V through the motor, through the transistor's collector and emitter to ground. When the button is released, the positive terminal node drops to 0V and the base is pulled low through the pull-down resistor, turning the transistor off and stopping the motor. A flyback diode is placed across the motor terminals, oriented from the negative (collector) side of the motor to the positive side, to dissipate voltage spikes generated by the motor's inductance when switching off.

"How it Works"

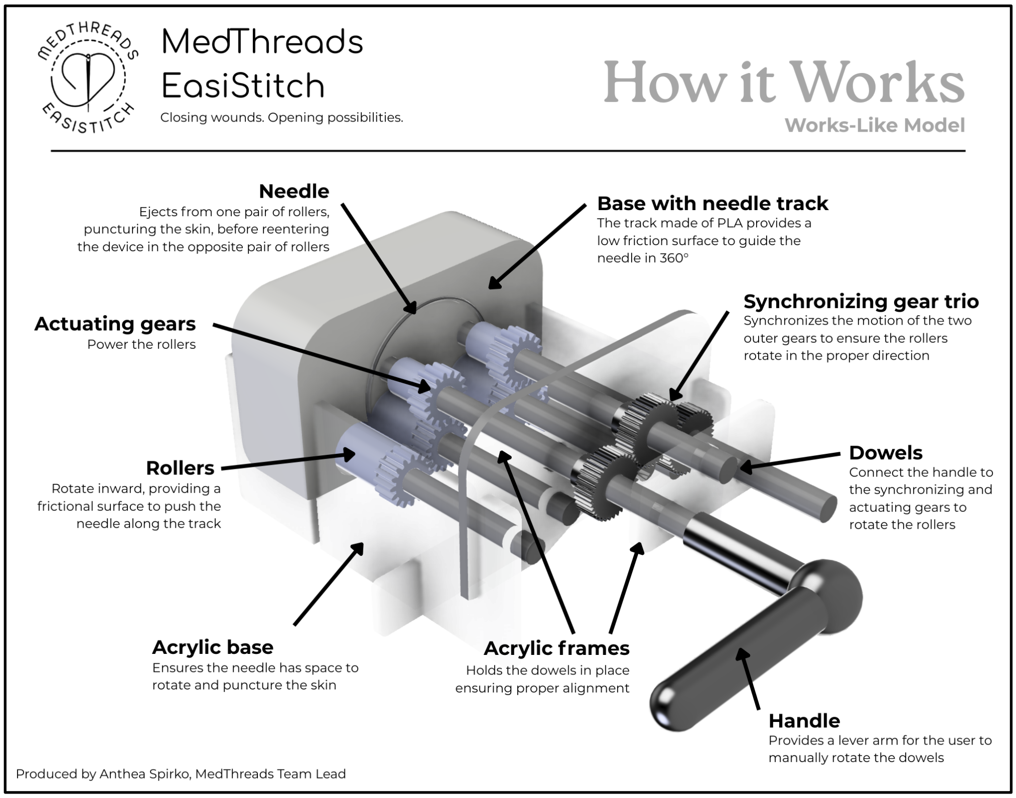

Figure 6. Works-Like Model How It Works Infographic

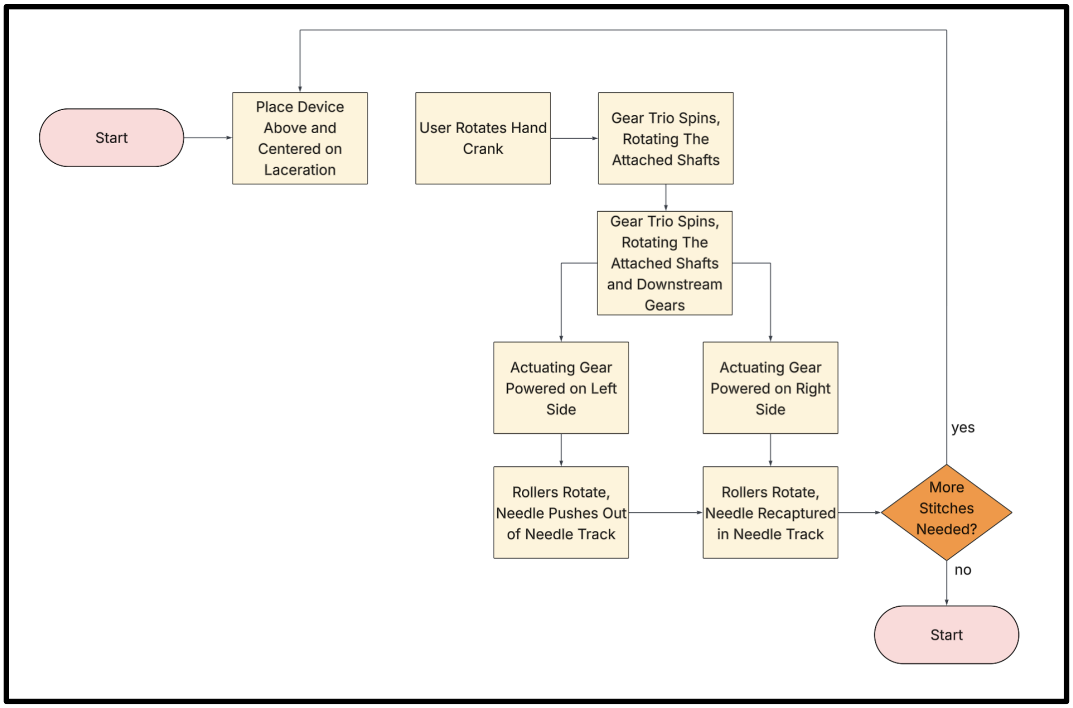

Figure 7. Work-Like Model High-Level Functional Block Diagram

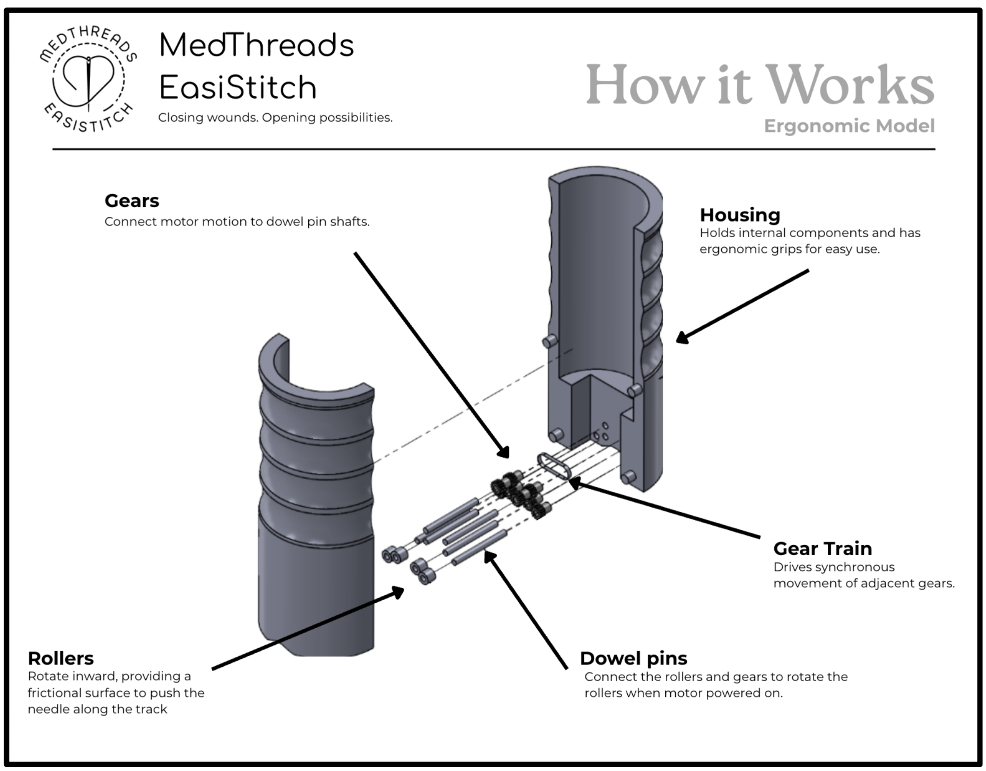

Figure 8. Ergonomic Model Internal Mechanical Components

Prototypes

User Interface

Works-Like Model

Given that the model is scaled up by a factor of 5, the mechanism is powered by user input via a hand crank rather than a motor and battery. For the user, the first interaction is placement of the device relative to the wound. Proper placement of the device is directly above the laceration and centered within the wound edges. The parallel surface of the device's base should interface with the skin on either side of the wound. The bottom of the device has a bottom inwards curve, helping facilitate the centering of the device along the wound. Once in place, the user must rotate the hand crank in a counterclockwise direction. This directionality is indicated by arrows next to the shaft with the hand crank. Once the user rotates the hand crank, the needle begins to travel along the needle track. After 360° motion of the needle, the user would move the device along the laceration to the next location of the desired suture. The output of each rotation would be the puncturing of the skin on either side of the wound, bringing the two edges together. Ultimately, the device will also contain a threading and cutting mechanism, delivering a suture along the needle track with controlled tension, depth, and bite.

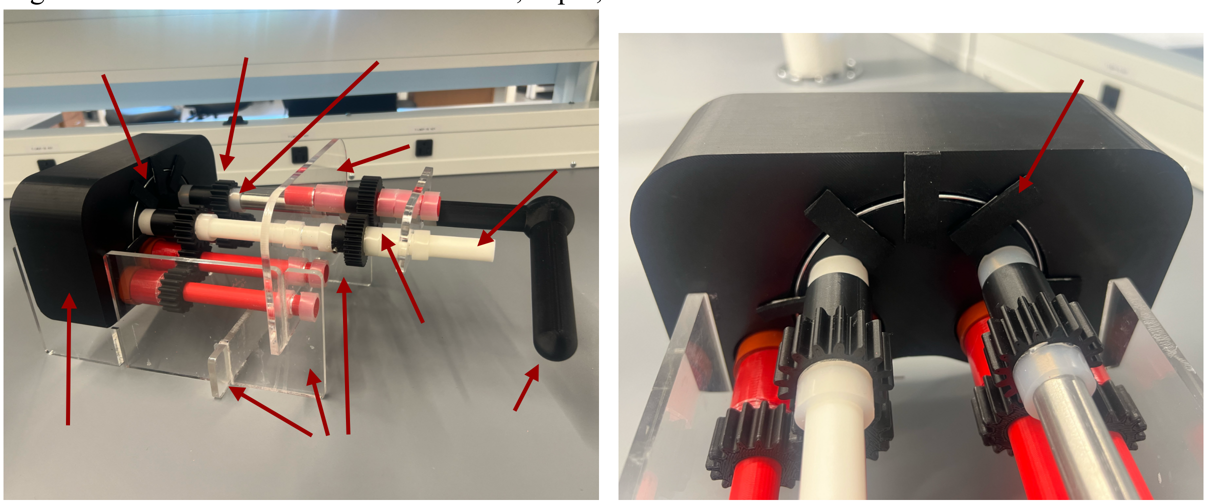

Figure 10. Work-Like Model

When prototyping this functional model, the main usability concern was ensuring that all components of the device are properly placed relative to each other. To address this, EasiStitch utilizes specially engineering acrylic frames. These frames are toleranced to ensure that the shaft is able to rotate freely. To then constrain the shafts within the frames, tubing was added on both ends. The gear trio, actuating gears, PLA base, and hand crank were all created within Fusion360 and printed using Black PLA Basic. Four of the six shafts were 3D printed using Red PLA Basic while the other two are dowels made of stainless steel and plastic. The needle here was bent using a 4-inch diameter cylinder and dremmeled to create a cutting end. To constrain the needle within the track, frictionless track covers were periodically spaced throughout. The rollers, which are embedded into the PLA base, were created using tape and a silicone overlaying sheet.

Validation & Verification

Overview

Both the proposed, 5x scaled-up internal mechanism and the handheld ergonomics of EasiStitich were tested. The functionality of both combined will be extrapolated via cumulative meta-analysis of both tests. The 5x mechanistic model of EasiStitch was powered via a hand crank of which the power generated is quite variable. Despite this drawback, the performed mechanistic test remains valuable due to resourcefully making the hand crank power in units of how many revolutions tools place. In other words, approximations of how many cycles of hand cranks equate meters (m) of movement to drive the enlarged simulatory needle through the track. The purpose of testing the scaled up prototype is to demonstrate the internal puncturing mechanism that would occur within our proposed cylindrical model. To validate this, the approximated trajectory of the puncture was visualized, along within the needle velocity and acceleration upon entry, exit, and throughout the needle's passage through the Dermasol skin model of 1 in thickness and 3.5 in diameter. Dermasol was chosen for the skin model due to it being transparent; meaning, it would allow for the digital tracking of needle movement. As such the Tracker app, a video analysis and modeling tool built on the Open Source Physics Java framework, was used to manually point-trace needle tip movement from entry and exit. Thereupon, the sequestered position, velocity, and acceleration data underwent cleansing and analysis in Python. The data of one completed stitch of the 5x model was theoretically compared to 3 wound closure trial runs across both stapling (n = 2) and suturing (n =1). Suture vs. staple times were also compared as well. All tests were performed within other Dermasol pancake models with all laceration simulations being about an 1 inch deep and 3.5 in length. The ergonomic model was tested using a survey of users to subjectively assess the hand-held experiences of EasiStitch for both wound closure efficacy and ease of use. Users were asked to hold and use both models and rate their poor to good (e.g., 1 to 5) experiences in handling the devices. Subsequently, those scores were visualized and compared. Data was analyzed with Google form and the backend Excel sheet.

EasiStitch Puncture Behavior

A 1 inch deep and 2.5 inch long gash was made to simulate a superficial laceration. The skin model was manually held into place to position it in the arc of the bite. To fully pierce the Dermasol pancake, it took about 4.5 revolutions of the hand-crank and 9.877 seconds for the needle to make a full path from ejection to being "caught" back into the frictionless track. The average acceleration was 0.2466 m/s^2, with a slightly lower average of 0.2205 m/s^2 during the first 2.125 seconds (Figure 12C). Average velocity overall was 0.0180 m/s, increasing slightly to 0.0190 m/s within the initial (Figure 12B) 2.125-second window. Between 7.3 and 7.7 seconds—the exit interval—the average velocity was 0.0185 m/s and the average acceleration was 0.1427 m/s^2. The higher acceleration accompanied by a slightly higher speed during initial skin contact and early penetration indicates the needle penetrated the softer, upper layers of the Dermasol with a continuous force. The slight decrease in both acceleration and velocity during the exit intervals may hint at an increased resistance or dermal drag from the deeper/denser layers of the skin. The overall estimated arc trajectory (Figure 12A and 13) is also somewhat comparable to the subdermal shape of the standard forceps driven suture needle (Figure 14). The manual suture needle was implanted perpendicular to a much thicker, in scale, skin model. Paired with the moderate continuous force applied to drive and catch the needle, it makes logical sense that the path would resemble a more ideal arc and, in turn, be visualized in the Dermasol skin model. However, the 5x model's scaled up needle was *propagated* through, rather than implanted within, the skin model. As a result, the suturing path did not follow an ideal arc likely due to the mechanics of the hand crank used to drive the needle (Figure 13). The crank probably introduced more variable force once the needle was inside the skin. In addition, while the entry and exit phases of the crank motion were relatively continuous, dermal drag during the internal phase likely compounded fluctuations in force and slowed the needle's progress. This effect was furthered by the condition of the skin model, which had been handled multiple times. This caused lateral sides to become opaque which, to a degree, reduced the visualization and tracing capability of the needle's path. Despite the drawbacks, the scaled model was able to complete a relatively smooth suture comparable path through a skin-like model. Indeed, EasiStitch's theoretical "release and catch" sewing mechanism functions securely and thus the mechanism built yields a feasible, continuous method of wound puncture.

Experimental Plots

<u>Path Analytics:</u>

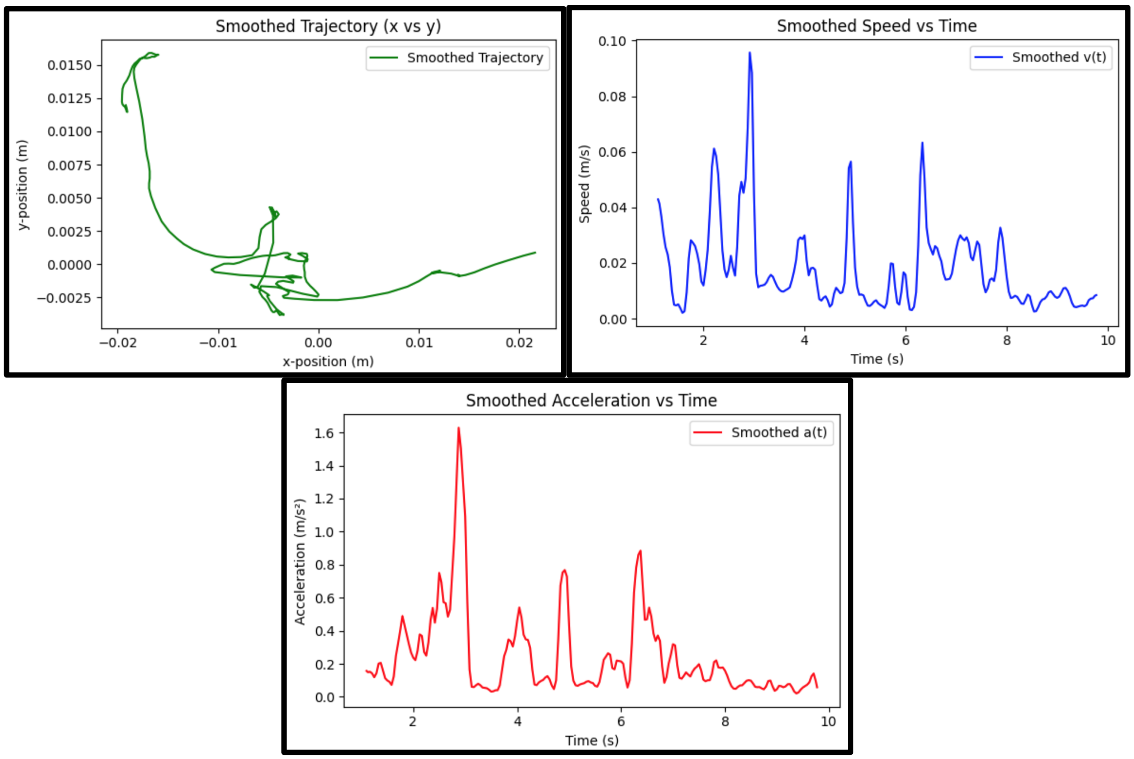

Figure 12. Needle Visualized Path Analytics: A) Trajectory B) Velocity C) Acceleration

Evaluation Path Shape:

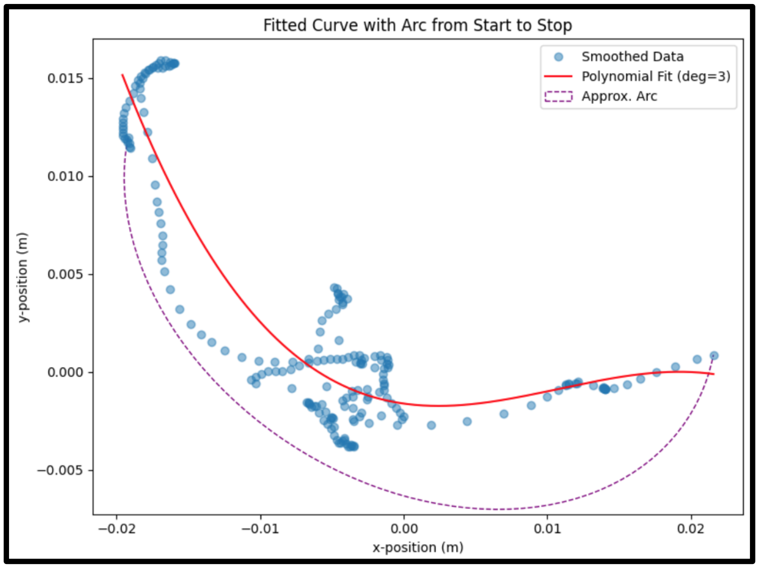

Figure 13. Ideal Arc (Purple) v. Path Taken of Needle (Red)

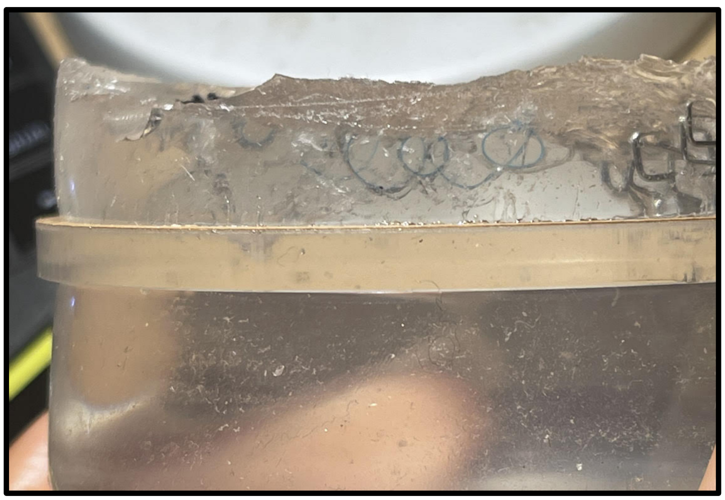

Figure 14. Side View of Dermasol from Suture Trial

Hand Suturing and Stapling Trials

Two trials of stapling a simulated laceration resulted in an average of 3.85 seconds per staple with 8 ejected staples during a 35.53 seconds during trial 1 or 4.44 seconds per staple (Figure 15A, right) and 13 ejected staples during a 42.31 seconds trial 2 or 3.25 seconds per staple (Figure 15B). The single trial of suturing (Figure 15A, left) took 12 minutes and 11 seconds for 8 suture or 1 minute and 30 seconds per suture. Expectations did include manual suturing to consume more time than staples. As the 5x scaled model took about 10 seconds to complete a closure, one can reasonably expect the scaled down, electrical model would *likely* puncture faster and, as such, perform closures faster than manual suturing. In addition, the electrical model would be able to more decisively propagate the needle through motor power to prevent dampening from dermal drag. However, it proves important to note that an amatuer, non-surgeon individual performed the manual suturing which may have contributed to the increased time consumed.

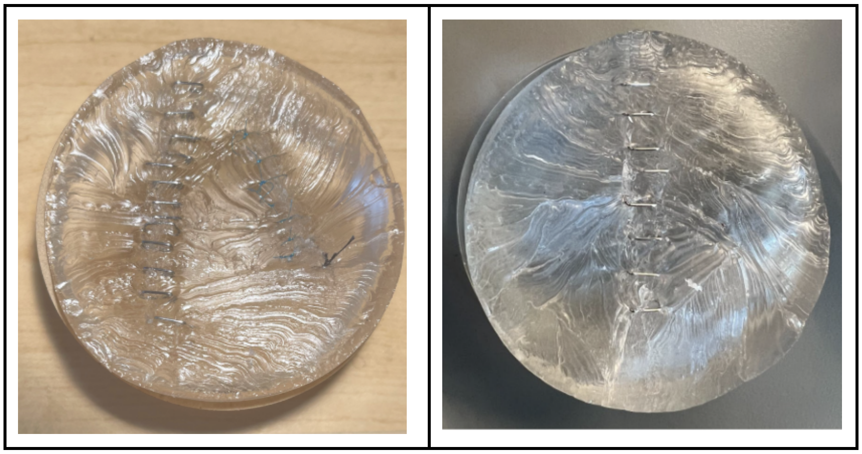

Figure 15. Post-Validation Samples:
A) Combined Staple Trial 1 (left) and Suture Trial (right) B) Staple Trial 2

User Testing

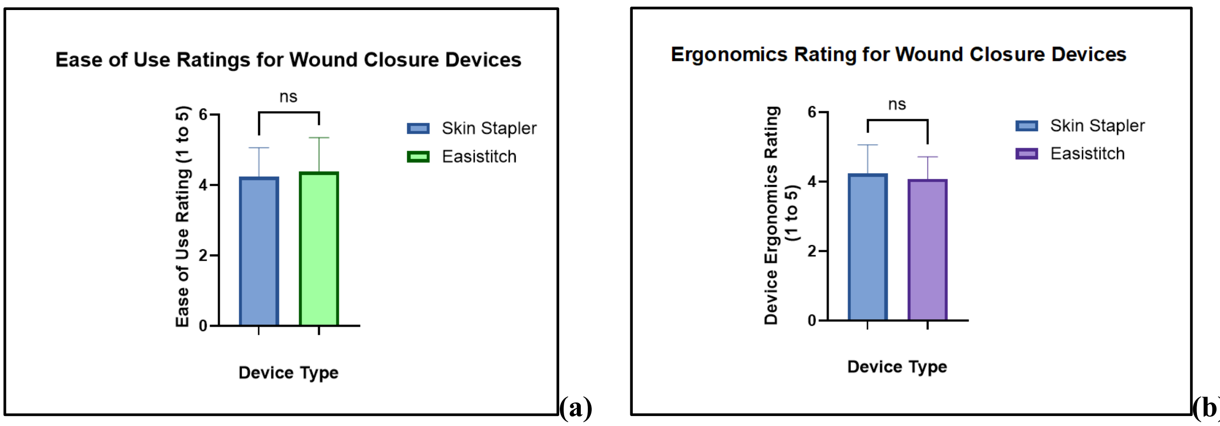

Figure 16. User Ratings Ease of Use and Ergonomics

The bar graphs show user ratings (n=10) comparing the skin stapler and our device, EasiStitch, on two usability aspects: ease of use **(a)** and ergonomics **(b)** on the panel above. In both categories, the skin stapler received slightly higher average ratings, but the differences were not statistically significant ($p > 0.05$). For ease of use, both devices scored around 4.3 to 4.5 out of 5, suggesting users found both devices similarly intuitive and simple to operate.

To evaluate our final design, we conducted a hands-on user study where participants interacted with a functional prototype of EasiStitch and a standard skin stapler. The lack of significant differences in scores suggests our design performs on par with the skin stapler. Users gave positive feedback overall, with some noting the grip could feel more comfortable. In response, we adjusted the handle shape to better fit the hand. These insights helped us validate the core usability of EasiStitch while fine-tuning smaller ergonomic details.

Differentiation

EasiStitch device stands out as a faster, more ergonomic, and more accessible solution to current wound closure methods. While manual suturing offers excellent precision and cosmetic outcomes, it is slow and requires high skill which greatly varies between physicians and medical personnel. Similarly, skin staplers are fast but often result in poor scarring and higher infection risk while adhesives are quick and painless methods of wound closure but are limited to clean and superficial wounds with no bleeding or tension. In contrast, EasiStitch provides consistent, cosmetically favorable sutures at a speed closer to that of staplers, with minimal user skill required. Existing automated or semi-automated suturing devices on the market (EndoStitch and QuickStitch) are designed almost exclusively for laparoscopic or endoscopic procedures and are too complex and expensive for routine external wound closure. Our device fills this critical gap by delivering reliable, uniform stitches through a handheld, intuitive system designed specifically for traumatic wound closure.

Potential Market

Potential market reach for our Easistitch device includes healthcare institutions such as hospitals, surgical centers, emergency rooms, and urgent care centers. End users of our device would include emergency medicine physicians, surgeons, and pediatricians who face time-sensitive emergencies and wound management scenarios on a daily basis. No components of Easistitch or even the device as a whole will be sold directly to the user.

In the United States, approximately 12.2 million surgical procedures are performed each year, and some sort of wound closure is required (Lewis & Pay, 2023). Targeting just 2.5% of all cases would represent approximately 300,000 potential units of our device to be sold in just the first year. When it comes to other wound closure medical devices, there are no direct competitors except traditional skin staplers and suturing kits. Other automated suturing devices, such as OverStitch or EndoStitch, are primarily used in endoscopic surgeries and thus do not overlap with our use case. Taking into account the unique combination of speed, precision, and cosmetic quality, MedThreads expects quick adoption of the Easistitch product in lower-resource settings where physician time is limited and access to resources is scarce. As we refine and scale up the production, we expect to reach approximately 10-15% of the market within the first 5 years.

Engineering Standards

The Easistitch development process was guided by several key engineering and medical device standards to ensure the success and safety of our device as well as plan for future regulatory requirements. ISO14971 was central in shaping our risk management approach and designing a safe electrical circuit. Current-limiting components, inductive load protection, and low-voltage components to minimize user and patient hazards. We also referenced our hazard analysis table and reconsidered the risks after hand calculating the max power/current and temperature changes of our final circuit in accordance with the standard's emphasis of proactive hazard analysis and mitigation. To prepare for manufacturing in a regulated environment, we ensured precautions were taken to have detailed documentation at each step of the design process including initial brainstorming, prototyping, and verification efforts. This helps align our workflow and future plans with ISO13485, defining the quality management requirements of medical devices. This would also be frequently referenced in order to implement full regulatory/quality tracking down the line, which would be needed throughout the regulatory approval process if we planned to further scale up our manufacturing and seek out FDA or regulatory approval of our product. Mechanically, we aimed towards and recommend the use ASTM D2256 for the tensile testing of our sutures, as was recommended to us by LSI solutions as the common standard used to perform a "Knot Pull-Apart" test of the strength of the sutures placed by our device. This would be important for testing that our device places and secures sutures without damaging or over tensioning the material, it would also important for internally validating the suture material itself which is purchased by external vendors rather than being produced in house.

As such, the compliance with engineering standards is important for not only medical device development but also regulation. Engineering standards ground and reinforce consistency, quality, and safety. EasiStitch's design and functionality must adhere to engineering standards documented in the ASTM (American Standards for Testing and Materials) section 13.01 (September 2025) and various other singular ISO standard codes. ASTM section 13.01 contains standards on medical and surgical materials that traverse many notable categories such as metals, polymers, and ceramics for implants, prostheses, and medical and surgical devices. In addition, other applicable ISO standards included 80601-2-58 (Medical electrical equipment), 10993 (Biocompatibility), 6335-2 (Surgical Instrument - Stapler), and 11608-1:2022 (Needle-Based Injection Systems) (ISO , n.d.). In addition, necessary FDA classifications include 21 CFR 878.4450 (Surgical Stapler) and 21 CFR 878.4750 (Surgical Suture with Needle) in which a 501(k) submission is necessary due to its automated nature.

Path to Clinical Use

To bring EasiStitch to market, several additional steps are required. Final R&D will focus on refining the needle actuation mechanism and validating consistent suture placement under different wound conditions. Device validation will assess tensile strength, suture eversion, and failure rates, while packaging and sterilization protocols must be validated to meet clinical standards. Usability testing in a simulated clinical environment, after obtaining IRB approval, will help demonstrate safety and effectiveness, especially because our design introduces new mechanics not found in current devices on the market.

Given that EasiStitch is a novel, automated Class II suturing device without a direct substitute, we anticipate pursuing a 510(k) regulatory pathway. The device does not meet the criteria for PMA since it is not considered high risk under FDA classifications, but the unique mechanism sets it apart from standard staples or suture kits. Through validation, usability studies, and clinical validation, we hope to demonstrate that the device is safe and effective.

EasiStitch will be distributed through institutional healthcare channels such as hospitals, surgical centers, and emergency departments. It will not be available over-the-counter in retail stores such as CVS or Walmart. Instead, it will be ordered in bulk by hospitals likely through partnerships with medical distributors. The device will be packaged as a sterile, single-use product and shipped in batches to clinical facilities. Long-term distribution will also involve clinical demos, physician training, and integration with hospital systems to support widespread adoption.

Selling Price & Reimbursement

The current unit cost of the EasiStitch device is approximately $251.70, primarily due to the use of SLA 3D printing (Tough 2000 resin), purchasing components in small batches (e.g., $19.55 per gear), and assembling the electronic circuit manually. However, such price is not reflective of the final cost of our device and MedThreads estimate that realistic per unit cost would be $75 dollars by assuming bulk volume manufacturing with injection molding and bulk order, optimized assembly processes, and vendor contracts for suturing and electronics equipment.

To maintain competitive edge on the market and justify unique features of the device, MedThreads proposes selling price around $150 to account for company revenue needed to pay employees and stakeholders. This estimate will allow for Easistitch to compete with current automated suturing devices such as OverStitch and EndoStitch which are being sold for over $500 dollars (*Endo Stitch Suturing Device, 3/Box*, 2023). Such a price range ensures that Easistitch remains affordable for hospitals while providing profit margins that would support future research and development in better iteration of the device.

Final Evaluation

Over the course of the semester, MedThreads focused on developing a novel-puncturing mechanism using a ½-circle needle, rotating in a 360° track. The needle motion is generated by two pairs of rollers actuated by a gear train. A pair of rollers ejects the needle while another pair recaptures the needle on the opposite side of the wound. Our device met our design criteria of maintaining consistent sutures by regulating the suture length, suture depth, and needle angle of entry. We were able to develop a scaled-up, 5X version of the gear setup to visualize the mechanism. This allowed us to identify points of difficulty in our design, which included the gear belt. We pivoted to use a gear trio to synchronize the pairs of rollers for the scaled-up version that is powered by a hand crank. This project was also successful in considering ergonomics. We were able to transfer the scaled-up gear-setup, to fit within an ergonomic, true-to-size model.

Due to time constraints of the semester, we had to delegate where we spent our time. This meant we were unable to fully flesh out the tying and cutting mechanism. As a team, we had the opportunity to visit LSI Solutions, located in Victor, NY. Here, we learned about some tying mechanisms, including crimping and automated knots, that we hope to extend to our device in later iterations. If we were to continue work on our device, we would have to have more modularity with our later prototypes, allowing for variations in suture and needle type, bite size, and bite depth. We were able to do preliminary testing to collect user input but would have liked to do more extensive testing had time allowed. As a collective, we developed our technical skills, including CAD, 3D printing, resin printing, electric circuit development, and general workshop skills. We learned that developing a scaled-up prototype can help solidify the mechanism before scaling it down again. More broadly, we collaborated with experts in the field to get into our design ideas, and save us time where we otherwise would have been reinventing the wheel.

APPENDIX: Engineering Analysis

In our final ergonomic design, we assume that the synchronizing gear trio rotates at the same speed as our high-torque motor. Given that the motor operates at 27 RPM, the synchronizing gears will also rotate at 27 RPM. To estimate the speed at which the needle enters the skin, we assume the needle motion directly matches the speed of the rollers. Since the rollers are driven by the actuating gears, and we assume no slippage occurs between components, the rollers and actuating gears will also rotate at the same speed. Therefore, the needle's entry speed is expected to match this rotational speed.

Knowing the number of teeth of actuating gears and synchronizing gear trio allows us to take advantage of the gear ratio in order to calculate the rotational speed of the rollers and needle.

$$\omega_{needle} = 27RPM \times \frac{16}{36} = 12\ RPM$$

Electrical safety was a key design consideration in the electrical development of the device. The measured steady-state current of the motor was 82mA, just under the 85mA load current listed on the datasheet. During normal operation with the supplied 12V from the batteries, this leads to an electrical power input of 1.02W (see all calculations below). When the button is pressed, the transistor's base voltage is determined by the voltage divider between the 470 ohm and 8.2kohm pulldown resistor. The chosen resistor values ensure functional switching behavior from the transistor and a flyback diode prevents voltage spikes due to inductance when the button is released. The chosen resistors result in a base voltage of 11.35V when the transistor is active.T he corresponding base current is around 24mA, which far exceeds the calculated 8.5 mA required to fully saturate the transistor for an 85mA collector current. Power dissipation across the active transistor is calculated to be only 60mW, which corresponds to a maximum temperature increase of 3.87˚C, making it reliably safe without the need for a heatsink as well. These values confirmed that the chosen resistor values and transistor are appropriate for the purposes of this device considering both functionality and safety.

$$P_{input} = 12V \times .085A = 1.02W$$

$$P_{disp} = .7V \times .085A = 59.5\ mW$$

$$V_{base} = 12V \times \frac{8200\ k\Omega}{470\Omega + 8200k\Omega} = 11.35V$$

$$I_{base} = \frac{(12V - .7V)}{470\Omega} = 24mA$$

$$I_{collector} = 12V \times \frac{8200\ k\Omega}{470\Omega + 8200k\Omega} = 85mA$$

Verify saturation with $\beta = 10$ (low to guarantee saturation): $I_{base} \geq I_{collector}/\beta \rightarrow 24mA \geq 8.5mA$

$$\Delta T_{max} = 59.5mW \times 65˚C/W = 3.87˚C$$

Figure 45. Ergonomic Model

One challenge that we faced was visualizing the gear mechanism, and perfecting the setup at such a small scale. Angela and Anthea began a pretotype, working on a scaled-up version. The initial model utilized foam rollers, 3D printed gears, and dowels found in the lab.

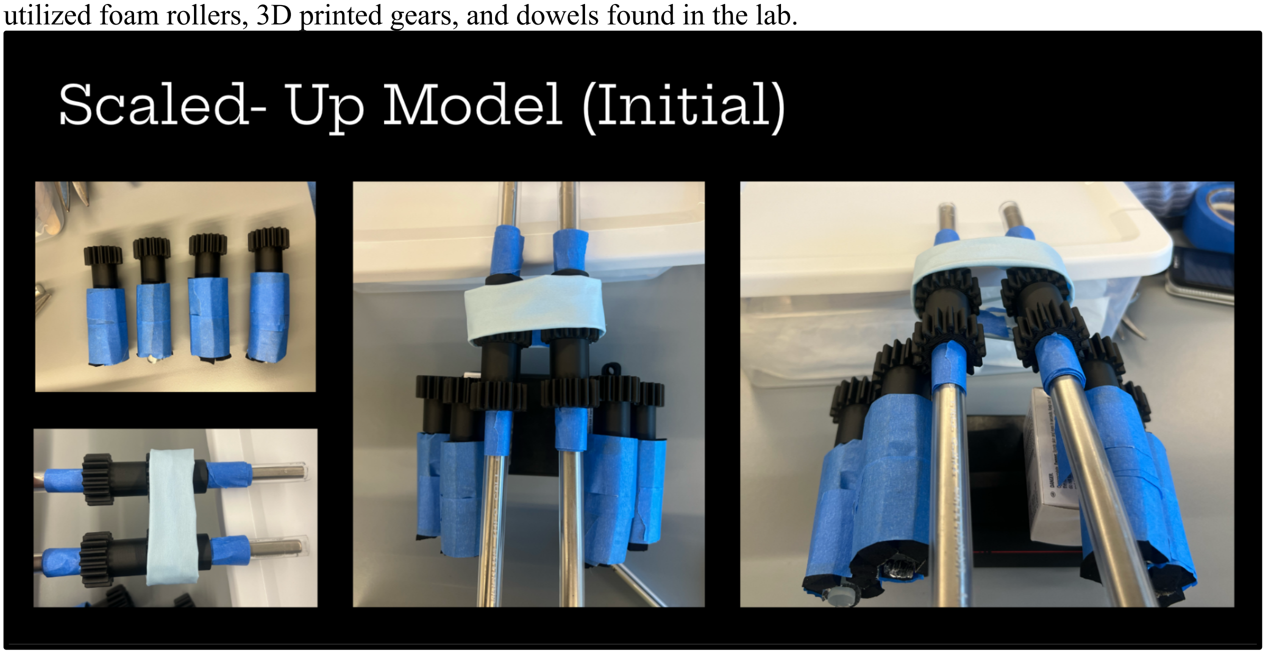

Figure 46. Initial 5X 'Works-Like' Model

This helped us to visualize the interfacing of the gears. From there, we began a CAD model of the scaled-up version. We laser cut an acrylic plate to stabilize the rollers and 3D printed the dowels, gears, and needle track. The 3D print stopped midway through the bottom base, but ended up working out because the needle track and the dowel holes were printed.